Jun-Kun Chen

Ph.D. Candidate in Computer Science | Expected Dec. 2026 | Seeking Research Scientist / Engineer roles from 2027

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Legal name: Junkun Chen | Publications: Jun-Kun Chen / J.-K. Chen

Research Focus

Controllable visual generation with diffusion models: spatially consistent 3D scene editing, temporally consistent video and 4D editing, multimodal control and minute-long human video, and now **world models that predict the visual outcome of actions**. First/co-first-author papers at NeurIPS 2025, NeurIPS 2024, CVPR 2024 (×2), 3DV 2026, and WACV 2027.

Industry Experience

Meta Reality Labs — **May–Dec. 2025; May–Aug. 2023**
Research Scientist Intern (2025 continued part-time Aug.–Dec.) — Bay Area, CA, U.S. / Zurich, Switzerland

- **2025:** Research on **joint 3D and video generation**, then video-to-3D and 3D-to-video generation; designed and implemented a **GPU-role-separated pipeline** for data processing and model training at **128–256-GPU scale** on Meta's distributed infrastructure.
- **2023:** *ConsistDreamer*: designed and built **self-supervised diffusion adaptation** and an async pipeline (**10× training speed-up**).

SpreeAI — **May 2024 – May 2025**
AI Research Intern — Remote, U.S.

- **Dress&Dance**: designed **target-steered supervision and staged training** for garment, identity, and motion control in video diffusion, generating 5-second try-on videos at **1152×720, 24 FPS**.
- **Virtual Fitting Room** (NeurIPS 2025): designed **reference-conditioned training from short-clip pairs**, enabling anchored autoregression for identity-consistent video up to **90 seconds**.

Selected Publications

1. Virtual Fitting Room: Generating Arbitrarily Long Videos of Virtual Try-On from a Single Image ↗ — **NeurIPS 2025**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Sep. 2025)

2. Dress&Dance: Dress up and Dance as You Like It ↗ — **WACV 2027**
Jun-Kun Chen, Aayush Bansal, Minh Phuoc Vo, Yu-Xiong Wang — Project | Early preview (arXiv, Aug. 2025)

3. ProEdit: Simple Progression is All You Need for High-Quality 3D Scene Editing ↗ — **NeurIPS 2024**
Jun-Kun Chen, Yu-Xiong Wang — Project | arXiv (Nov. 2024)

4. ConsistDreamer: 3D-Consistent 2D Diffusion for High-Fidelity Scene Editing ↗ — **CVPR 2024**
Jun-Kun Chen, Samuel Rota Bulò, Norman Müller, Lorenzo Porzi, Peter Kontschieder, Yu-Xiong Wang — Project | arXiv (Jun. 2024)

5. Instruct 4D-to-4D: Editing 4D Scenes as Pseudo-3D Scenes Using 2D Diffusion ↗ — **CVPR 2024**
Linzhan Mou*, **Jun-Kun Chen***, Yu-Xiong Wang — Project | arXiv (Jun. 2024)

6. HDEdit: Editing Videos and 3D Scenes with Video Diffusion Through Hierarchical Task Decomposition ↗ — **3DV 2026**
Yanming Zhang*, **Jun-Kun Chen***, Jipeng Lyu, Yu-Xiong Wang — Project | Earlier version (arXiv, Mar. 2025)

*Equal contribution. Earlier publications also appear as Junkun Chen / J. Chen.

Current Research: Generative World Models

Generative world models that predict what happens next on a screen, in a physical scene, or under a robot's motion.

Computer Use — Next-Screen Image-Diffusion World Model (2025–26, lead, mentoring a junior student)

- Built a next-screen world model on SenseNova MoT: a language Reasoner describes the action's effects, and a Generator adapted by **diffusion post-training** renders the next screen.
- Formulated supervision from a **teacher that sees the real next screen** (what changed, what must stay) to train the Reasoner.

Physics — Video-Diffusion World Model for Physical Events (2025–26, lead, mentoring a junior student)

- Adapted **Cosmos3 video diffusion** to predict physical events from past RGB frames and structured event descriptions.
- Implemented **teacher distillation** from a full-video teacher to a Reasoner that sees only past frames; 4-GPU FSDP2 training.

Robotics — Motion-Conditioned Video Diffusion World Model (2025–26, solo project)

- Trained a **motion-conditioned video world model** on simulated and real robot data (MimicGen, RoboTwin, DROID, RH20T, AgiBot) to predict interaction video from a scene, recent frames, and rendered robot motion.
- Conditions on a **rendered robot-motion proxy plus paired motion–video examples**; studying adaptation to new setups.

Education

University of Illinois Urbana-Champaign — **2021 – Dec. 2026 (expected)**
Ph.D. Candidate in Computer Science; Advisor: Prof. **Yu-Xiong Wang** — Urbana, IL

Tsinghua University — **2017 – 2021**
B.Eng. in Computer Science, Yao Class (IIIS); Major GPA: 3.94/4.0 — Beijing, China

Technical Skills & Honors

Tools & Methods: PyTorch; Python; C++; FSDP2; Slurm; image/video diffusion (DiT, rectified flow); LoRA; VLMs; diffusion post-training (SFT, DiffusionNFT); teacher distillation; human/VLM-judge evaluation; paired-bootstrap statistics.
Competitions: ICPC EC-Final Gold (4th); ICPC Qingdao Gold (2nd); CCPC Harbin Gold; NOI Gold; NOI/ICPC problem setter.

Selected Research Contributions

Learning Visual Control and Long-Horizon Generation — Dress&Dance and VFR

- **Dress&Dance:** designed **target-steered supervision and staged curricula** for new visual controls; FVD **157 vs. 197** (Internet) and **20 vs. 49** (captured), ours first, lower is better, vs. a Kling 1.6 try-on-and-animation pipeline.
- Designed **CondNet**, one conditioning space for text, layout, pose, garment, and identity, plus **synthetic-condition/real-target triplets**; hybrid image/video training and progressive schedules add controls while retaining pretrained priors.
- **Virtual Fitting Room:** designed **short-clip pair sampling** to learn reference-conditioned appearance and prefix continuation, enabling generated 360° anchors at inference for identity-consistent video up to **90 seconds**; on 30–50 s hard-motion tests, highest VBench subject/background consistency and motion smoothness (**93.2 / 95.3 / 99.4**) vs. D&D+FramePack and D&D+Kling 2.0, with comparable GPT-judged try-on quality.

General-Purpose Diffusion Editing — HDEdit and ProEdit

- **HDEdit:** designed **hierarchical task decomposition** that separates instruction following from content preservation for training-free video and scene editing.
- **ProEdit:** designed and implemented tuning-free preservation control and a **progressive editing workflow** with previews and edit-strength control for high-fidelity geometry and appearance editing.

Multi-View and Temporal Consistency — ConsistDreamer and Instruct 4D-to-4D

- **ConsistDreamer:** designed **cross-view self-supervision and staged diffusion adaptation** using warped-view targets, geometry-aware ControlNet/LoRA, and consistency losses across the **denoising trajectory**.
- Implemented a **4-GPU async pipeline** overlapping diffusion training, generation, and scene fitting (**10× training speed-up**).
- **Instruct 4D-to-4D:** extended 2D diffusion to long-term dynamic scenes through **anchor-aware attention** and optical-flow-guided sliding windows.

Geometric Visual Representations — NeuralEditor and PointTree

- Designed point-based NeRFs and relaxed k-d trees for editable geometry and **transformation-robust point-cloud encoding**.

Additional Publications

7. SceneCraft: Layout-Guided 3D Scene Generation ↗ **NeurIPS 2024**
8. Contrastive Learning Relies More on Spatial Inductive Bias Than Supervised Learning: An Empirical Study ↗ **ICCV 2023**
9. NeuralEditor: Editing Neural Radiance Fields via Manipulating Point Clouds ↗ **CVPR 2023**
10. PointTree: Transformation-Robust Point Cloud Encoder with Relaxed K-D Trees ↗ **ECCV 2022**
11. Is Self-Supervised Contrastive Learning More Robust Than Supervised Learning? ↗ **ICML 2022 Workshop**
12. TorchDrug: A Powerful and Flexible Machine Learning Platform for Drug Discovery ↗ **arXiv preprint, 2022**
13. RNNLogic: Learning Logic Rules for Reasoning on Knowledge Graphs ↗ **ICLR 2021**
14. Pre-training Assisted Scene-aware Dialogue Generation ↗ **DSTC8 Workshop at AAAI 2020**

Earlier Experience

Mila – Quebec AI Institute **Feb. – Aug. 2020**
Research Intern; knowledge-graph reasoning with Prof. Jian Tang Montreal, Canada

Baidu, Natural Language Processing Department **July 2019 – Jan. 2020**
Research / Development Intern; scene-aware dialogue generation (DSTC8 top-3 system) Beijing, China

Competitions, Leadership & Coursework

Programming Competitions: ICPC EC-Final 2018, Gold Medal (4th); ICPC Qingdao Regional 2018, Gold Medal (2nd); CCPC Harbin Regional 2017, Gold Medal; NOI 2016, Gold Medal.
Additional Distinctions: CTSC 2017 (7th); CTSC 2016 Gold (8th); NOI Winter Camp 2016 Gold (3rd).
Leadership & Problem Setting: Set 9 contest problems (2018–2019), including hardest-tier tasks at NOI 2018 and NOI 2019, plus problems for CTSC (China's IOI team selection), NOI Winter Camp, THUPC (Tsinghua University Programming Contest), ICPC Hong Kong, and CCPC Qinhuangdao. Vice Chair, Tsinghua Student Association of Algorithms and Competitions, 2018–2019.
Coursework (Generative Models & Learning): Deep Generative Models (A+); Generative Models for Biomedicine and Life Sciences (A); Meta-Learning (A+); Machine Learning (A).
Vision & 3D: 3D Vision (A+); Deep Learning for Computer Vision (A); Computer Vision (A); Computational Photography (A+).